

2.6 Worksheet

1. Approximate the integral $\int_0^3 \sqrt{x+1} dx$ using...

a) a midpoint approximation with $n = 6$. Round your answer to two decimals.

b) a trapezoidal approximation with $n = 3$. Round your answer to two decimals.

c) Simpson's Rule using $n = 6$. Round your answer to two decimals.

2. Consider the integral $\int_2^4 \frac{1}{x^3} dx$.

a) Approximate the value of this integral using a midpoint approximation with 4 rectangles. Round your answer to four decimals.

i. Determine an upper bound on the error of this approximation using the error estimation formula discussed in the notes. Round this upper bound to 2 places and interpret this value.

ii. How large must n be to ensure an error of no more than $\frac{1}{80,000} = 0.0000125$?

b) Approximate the value of this integral using a trapezoidal approximation with 4 trapezoids. Round your answer to four decimals.

i. Determine an upper bound on the error of this approximation using the error estimation formula discussed in the notes. Round this upper bound to 2 places and interpret this value.

ii. How large must n be to ensure an error of no more than $\frac{1}{80000} = 0.0000125$?

- c) Approximate the value of this integral using Simpson's Rule with $n = 4$. Round your answer to two decimals.
- i. Determine an upper bound on the error of this approximation using the error estimation formula discussed in the notes. Round this upper bound to 2 places and interpret this value.
- ii. How large must n be to ensure an error of no more than $\frac{1}{80000} = 0.0000125$?

3. Consider the integral $\int_0^{\pi/2} \cos(x) dx$.

a) Approximate the value of this integral using a midpoint approximation with 3 rectangles. Round your answer to two decimals.

i. Determine an upper bound on the error of this approximation using the error estimation formula discussed in the notes. Round this upper bound to 2 places and interpret this value.

ii. How large must n be to ensure an error of no more than 0.0001?

b) Approximate the value of this integral using a trapezoidal approximation with 3 trapezoids. Round your answer to two decimals.

i. Determine an upper bound on the error of this approximation using the error estimation formula discussed in the notes. Round this upper bound to 2 places and interpret this value.

ii. How large must n be to ensure an error of no more than 0.0005?

- c) Approximate the value of this integral using Simpson's Rule with $n = 4$. Round your answer to two decimals.
- i. Determine an upper bound on the error of this approximation using the error estimation formula discussed in the notes. Round this upper bound to 2 places and interpret this value.
- ii. How large must n be to ensure an error of no more than $\frac{1}{100}$?